

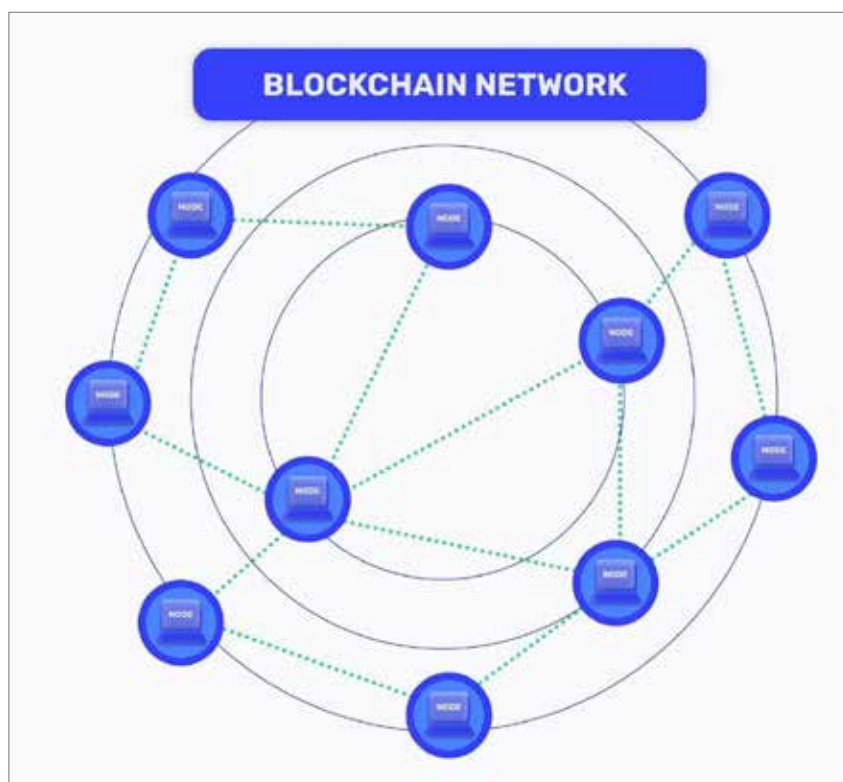


Figure 1: A simplified blockchain network with interconnected nodes maintaining the distributed ledger

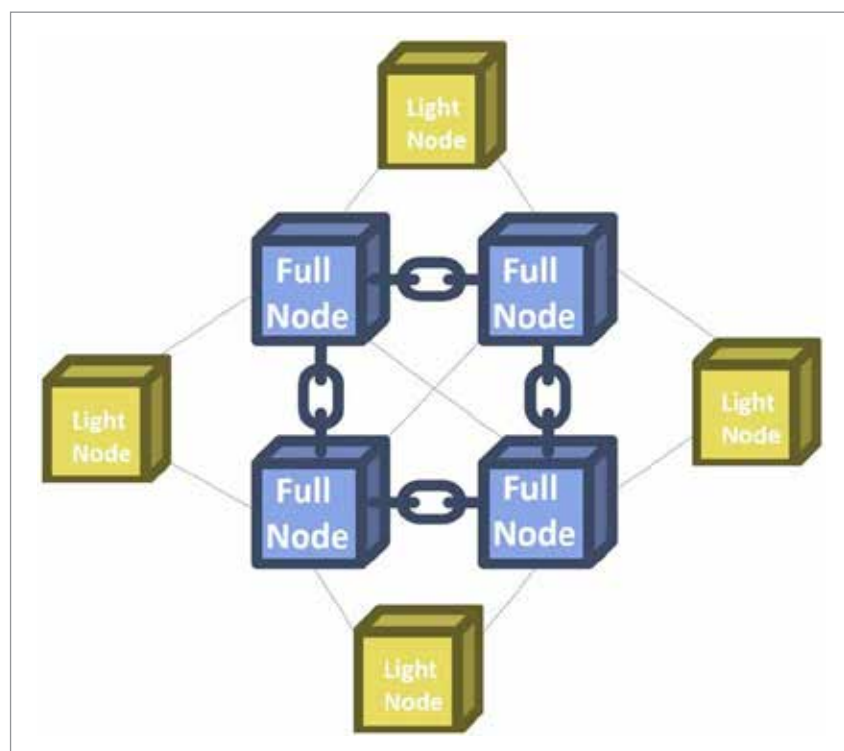


Figure 2: Full nodes and light nodes interacting within a blockchain network

restarts, it needs to sync with the blockchain. This process can take significant time, especially in large networks. If not managed properly, this can lead to downtime or inconsistent system behaviour.

In addition, the nodes must be connected to other peers in the network. Network instability or configuration issues could cause nodes to fall out of sync or lose connectivity, which would affect their ability to validate and share the data.

Because of these factors, managing blockchain nodes is about maintaining a distributed system that is reliable, scalable, and is monitored continuously.

Challenges in running blockchain nodes

While blockchain networks are designed to be decentralised and resilient, running the underlying nodes in production could introduce several practical challenges, the biggest one being storage. Blockchain data keeps growing continuously, and a full node is required to store the entire chain. Over time, this can reach hundreds of gigabytes, or even more, depending on the network. Managing this data, ensuring sufficient disk space, and handling backups become critical.

Another issue is synchronisation time. When a node is started for the first time or restarted after failure, it needs to be synced with the network. This process could take several hours or even days in case of larger networks. During this time, the node is not fully usable, which can affect system availability.

Node failures are also common in real-world environments. Hardware issues, network interruptions, or software crashes can cause nodes to go down. Recovering from these failures manually often requires intervention, which increases operational overload.

The other concern is about configuration management. When multiple nodes are deployed across various environments, keeping